# Learning Cancer-Related Drug Efficacy Exploiting Consensus in Coordinated Motility Within Cell Clusters

Davide Di Giuseppe, Francesca Corsi, Arianna Mencattini, Maria Colomba Comes, Paola Casti, Corrado Di Natale, Lina Ghibelli, and Eugenio Martinelli

***Abstract*—*Objective:*** **The ability of cells to collectively move is essential in various biological contexts including cancer metastasis. In this paper, we propose an automatic video analysis tool to correlate the cell movement inhibition with replication block induced by dose-dependent chemotherapy administration.** ***Methods:*** **The novel approach combines individual and collective cell kinematic analysis performed over time-lapse microscopy video frames. Cells are first localized and tracked, and then kinematic descriptors are extracted for each track. Selective track identification is performed assuming diversified cell roles within the same cluster (spontaneously forming groups of cells), and finally individual results are grouped exploiting consensus of coordinated motility within cell clusters.** ***Results:*** **Recognition performance of three different experimental conditions (no drug, 0.5–5 $\mu$M merged in the same condition, and 50 $\mu$M) reached an average accuracy value of 88% over 958 different tracks collected in 36 clusters of diverse dimensions in eight independent experiments.** ***Conclusion:*** **An extensive application of this methodology could give a different point of view of the cancer mechanisms.**

***Index Terms*—Cell tracking, cell motility, cancer cell replication block, multivariate data analysis.**

## I. Introduction

THE ability of cells to coordinate their motion is essential in various biological contexts, including morphogenesis [1], tissue repair [2], [3], cancer progression and metastasis [4]. In most examples, collective motility is not simply the result of many individually moving cells: cells crawling together produce unusual behavior for individual cells such as emergent chemotaxis [5] or cell leader/follower paradigmatic roles in movement coordination [6]. Collective movement is however a very complex process including different elements [7]: the motility of single cells within the collective to represent intrinsic cell heterogeneity due also to cell-intrinsic states, the representation of cell shape and consequent mechanical interactions, the cell-polarity mechanism (direction where cell would like to move in absence of external constraints), and not less relevant, the potential biochemical signaling between cells. Development of closed-form models for such complexity may force to simplified assumptions and may produce specific or simplistic cell models [8]. On the other hand, experimentally-driven intervention to alter motion mechanism (i.e., replication as well as cell motion inhibition by chemotherapy) influences several features at once, and may lead to ambiguity in human-interpretation of data. Standard approaches to the problem of quantitatively investigate cell motility involve the extraction of a unique model for all the cells in a cluster under the assumption of a commensurate role in the cluster. On the other hand, attempts to define an individual role for each cell involved may lead to very complex parametric models difficult to be optimized and applied. Violation of these assumptions is one of the main characteristics of this work.

The alternative solution proposed in this work involves a two-fold paradigm. First, individual cell movements are accurately tracked, characterized in terms of kinematics descriptors collected over time, and each track is used to train machine learning algorithms. As suggested in the literature [7], this step is required to cope with cell heterogeneity due to internal states, polarization, cell diverse roles [9], [10]. However, the cluster state deserves the highest attention in regards to the time-varying dose-dependent drug-efficacy evaluation. To this aim, a consensus rule is then applied to the results achieved on individual tracks in order to provide a global quantification of drug impact on cluster motility inhibition, i.e., drug efficacy. To demonstrate the feasibility of the proposed architecture, videos of cancer cells moving in Petri dishes are acquired at 1 frame per minute with a resolution of 0.33 $\mu$m/px. The collected images are then processed through the *Cell-Hunter* software [11], [12]. Cell-hunter plays a crucial role in the platform since it allows to effectively locate and track cells over the video sequence without any human intervention. Cell-Hunter has been selected for its already proven capability to achieve reliable results in the very critical biological context of immune-tumor microenvironment

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[11], [12]. Not least, it requires a small set of parameters to be set, being almost prevalently adaptive, and includes localization and tracking modules specifically tailored for circular-shaped cells. Anyway, to further validate its effectiveness, Cell-Hunter tracking results have been compared with those obtained by the publicly available software *ImageJ* [13] demonstrating increased robustness and reliability. Cell tracks extracted through the use of Cell-Hunter are then coded into a set of kinematics descriptors. Such parameters concur to multivariate analysis of coordinated motility within the cluster of cells with the aim to globally explain cell behavior under different biological conditions.

### A. Related Work

First attempts to automate the tracking of particles by digital image processing date back at least 30 years [14]. Since then, the development of more advanced tracking methods has progressed enormously but it is only during the latest years that biology has practically enjoyed these efforts through the increased availability of software implementations [14], [15]. If from one hand a software tool provides support to biologists in correcting and reducing artefacts due to laborious error-prone process of manual video reviewing [16], it may also open the frontiers to new challenging scenarios in time-lapse microscopy: to analyze cell migration by automatically tracking high-dense particle populations avoiding invasive fluorescence-based labelling technique. *Cell_Motility* software [17] implements individual cell-migration analysis starting from already acquired trajectories. Hence, time-consuming manual tracking biased the validity of the analysis. *Pathfinder* software [18], [19] focuses on migration of clusters of cells on the assumption of the existence of a unique representative group migration pattern for a specific experiment. Pathfinder is based on the well-founded tracker *ParticleTracker* [19], a tool developed for particles detection and tracking in fluorescent images. Pathfinder violates the assumption of cell motility individuality (internal state, polarization, diverse role) and cannot be used in experiments where cell heterogeneity plays a crucial role in the collective migration process. *AveMap* tool [20] is based on the particle image velocimetry (PIV) technique and consists of dynamically mapping the velocity field within the monolayer under the key assumption of confluent migration of investigated cells. However, in the observed experiments, the collective motion is qualitatively represented by a combination of disordered motion and states of coherent angular motion [21] thus reducing potential applicability of PIV techniques. To authors' knowledge, *Trackmate*, [13] a plugin of Fiji distribution of the ImageJ tool, can represent a promising tracking tool, obtaining the best performance when applied to real images. Unfortunately, only few attempts can be found in order to combine tracking tool with cell descriptors [22]–[24]. However, for the sake of results validity, as mentioned before, analysis has been carried on by comparing Cell-Hunter and ImageJ. Results achieved demonstrated comparability of results in favor of an improved versatility of Cell-Hunter in terms of reduced set of parameters and specifically tailored localization and tracking modules.

### B. Main Contribution

The original aim of this study is to introduce and validate a novel automated cell kinematic analysis including different machine learning and data analysis modules, such as automatic cell tracking, kinematic features extraction, classification based on a majority voting reasoning. To this regard, in order to demonstrate the feasibility and the effectiveness of the proposed approach, we selected a "real" bio-system environment consisting of cancer cells (PC3 in this case) treated with a chemotherapeutic drug (etoposide). More specifically, since cell replication and motility are associated parameters in cancer progression, we wanted to investigate if blocking replication automatically affected motility. We thus decided to focus on two aspects: the data analysis approach (to our knowledge, such question had not been previously addressed in the relevant literature) and the first evidence of a novel cell-based motility biosensor.

More in details, main novelties in the proposed platform rely on the ability to combine individual and collective cell analysis by locating and tracking cells, perform individual track analysis, apply selective track identification, and finally group the results achieved over the same cell cluster by a majority voting procedure. One of the crucial assumption embedded in the proposed model is the diverse role of cells according to the position in the clusters [10], [21]. Cells belonging to the core seem to be less representative in motility variation effects than cells on the cluster rim. Such strategy allows the platform to collect a huge number of data for each experimental condition (e.g., treated vs untreated) thus assuring statistical significance of the results while maintaining the intrinsic cell heterogeneity fundamental for the detailed description of the underlying biological mechanism. This ability overcomes the less general assumption of cell-based drug effect isotropy (e.g., should all cells have been influenced by the same amount of drug in exactly the same manner?) that is directly connected to the assumed presence of diverse cell-roles (leader/follower) in collective migration [6]. On the other hand, the proposed methodology assumes the presence of a global cluster behavior not in terms of cluster migration descriptors but as drug-motility inhibition effect. This result is obtained by computing the majority result assigned to all the cells in the same cluster. To do this, a multivariate data analysis model has been used to combine descriptors extracted from different cell tracks in a unified kinematic model of drug-effect. The platform can be then used to evaluate drug efficacy as well as related cell/cluster isotropic effects in a given sample.

## II. Material and Methods

### A. Materials

PC-3 prostate cancer cells were grown at 37 °C in RPMI 1640 medium supplemented with 10% fetal bovine serum FBS (Gibco), 2 mg/mL L-glutamine, 100 IU/mL penicillin and streptomycin (Euroclone) in a humidified atmosphere of 5% $CO_2$ in air. In all experiments, cells were seeded in 35 mm petri dishes (Greiner) and grown to 40–60% confluence. Cells were treated with the topoisomerase II inhibitor etoposide (chemotherapeutic drug, Sigma) at the final concentrations of 0.5 $\mu$M, 5 $\mu$M or

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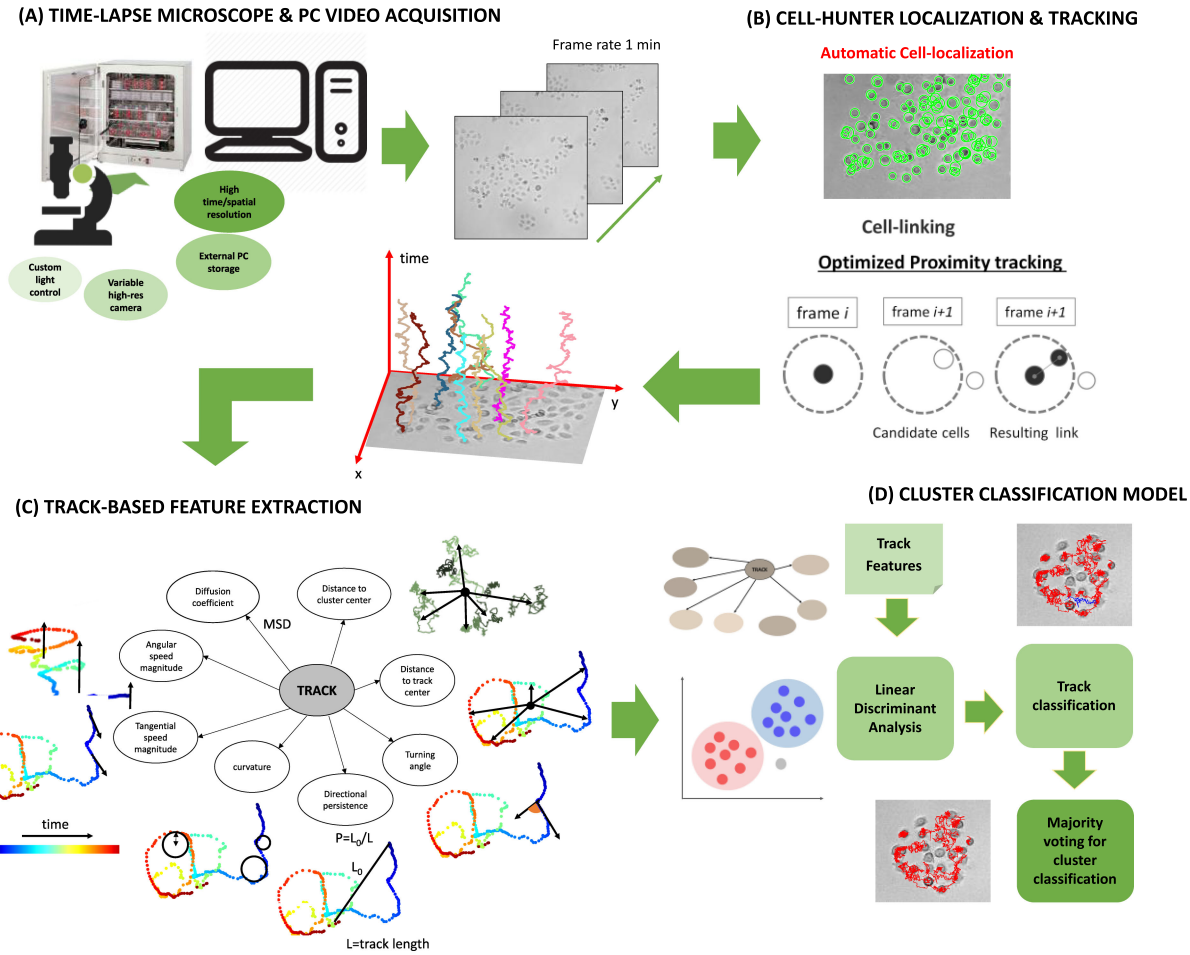



Fig. 1. A schematic view of the whole architecture. (A) Time-lapse microscopy of the experiment and video acquisition. (B) Cell localization and tracking (cell linking). (C) Track-based feature extraction for cell kinematics analysis. (D) Cluster classification model.

50 $\mu$M and immediately analyzed for time lapse for the indicated times.

### B. A Novel Platform for Analysis

Figure 1 summarizes all the main blocks involved in the architecture. First, panel (A), a customized microscope has been implemented: the prototype consists of a small-scale inverted microscope suitable to work in high humidity environments (incubators) with factory standard optics, custom aluminum structure and fully sealed electronics. The whole microscope dimensions are less than 400 mm in height and 350 mm in length and depth. Furthermore, a custom firmware was implemented in Matlab2017a environment in order to have full control on acquisition methods and light exposure. In this work, we acquired at one frame per minute, with 6 h of total experiment time. The captured images have a FOV of 1.2 mm width by 1 mm height with a spatial resolution of 0.33 $\mu$m/px. Second, panel (B) the acquired video is processed by *Cell-Hunter* software [11], [12] able to automatically locate population of cells in each video frame and then perform an optimized linking procedure between consecutive frames based on the concept of cell proximity and optimal assignment problem [25], [26]. Each track is then characterized by using high-level statistical features computed on the set of kinematics descriptors pictured in panel (C). Each track is then coded in terms of its motility characteristic in time and space domain and send to the classification model block (panel (D)). Here, a golden standard classification model as Linear Discriminant Analysis (LDA) is implemented in order to assign a label to each track, according to a predetermined list of possibilities used during the training of the model. The underlying assumption of our study is that a cluster, still preserving its own internal heterogeneity, carries on a unified biological message (i.e., a unique label for drug efficacy): under this hypothesis, we implemented a majority voting procedure over the labels assigned to the tracks of the same cluster, so that the final labels provided in a given experiment can be used as a signature of drug effect.

***1) Cell-Hunter Localization and Tracking:*** Automatic localization of cells in each video frame is the fundamental step in any automatic tool for cell-cycle monitoring. High-frame rate time-lapse microscopy (i.e., without fluorescence effect) can present challenging tasks in cell localization especially related to cell-overlapping, partial cell occlusion, variable number of cells in the video. Standard Otsu's thresholding and/or region-based segmentation approaches [27] fail due to the aforementioned problems. Cell-Hunter tool implements

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the Circular Hough Transform (CHT) [11], [12], [28] localization/segmentation method, under the assumption of circular-shaped objects, that simultaneously allows to locate cell nuclei in a given image providing an accurate estimate of individual cell radii. Cell occlusion and overlapping problems are solved through the geometrical mechanism behind. After cell nuclei are located, an improved proximity tracking approach based on the Munkres' solution to the non-square Optimal Assignment (OA) problem is applied. Track refining procedure related to the reliable assumption of low expected cell motility with respect to the very high frame rate achieved by the microscope allows to reduce false tracks and join small tracks in a whole.

***2) Track Kinematics Analysis:*** Provided the coordinates of a track, $(x(\mathrm{t_k}), y(\mathrm{t_k}))$, for $\mathrm{t_k} = \mathrm{t_0} \ldots \mathrm{t_f}$, with $\mathrm{t_0}$ and $\mathrm{t_f}$ the starting and the ending time instant of the track, the first step of the analysis is to extract kinematic descriptors characterizing the motility behavior of the track. First, to assure the validity of the extracted descriptors, we eliminated tracks with a duration smaller than two hours. Such tracks can result from tracking errors for example due to total cell occlusion during video. Then, a smoothing spline approximation is constructed for each track in order to reduce meaningless change of movements due to side effects, leading to the novel set of coordinates, $(x_s(\mathrm{t_k}), y_s(\mathrm{t_k}))$ for a trajectory, $\tau(x, y)$. After that, a set of standard parameters [29] have been computed as follows.

i) Tangential speed magnitude, $\mathrm{v}(\mathrm{t_k})$, given by

$$v\left(t_k\right)=\sqrt{\left(\frac{x_s\left(t_{k+1}\right)-x_s\left(t_k\right)}{\left(t_{k+1}-t_k\right)}\right)^2+\left(\frac{y_s\left(t_{k+1}\right)-y_s\left(t_k\right)}{\left(t_{k+1}-t_k\right)}\right)^2}. \tag{1}$$

ii) Track curvature, $\kappa(t_k)$ in $(x_s(\mathrm{t_k}), y_s(\mathrm{t_k}))$, given by

$$\kappa\left(t_k\right)=\frac{\left|x'_s y''_s-y'_s x''_s\right|}{\left[\left(x'_s\right)^2+\left(y'_s\right)^2\right]^{3/2}}, \tag{2}$$

where $x'_s$, $y'_s$ and $x''_s$, $y''_s$ are the first and second temporal derivatives, respectively, computed by finite differences.

iii) Angular speed magnitude, $\omega(\mathrm{t_k})$, computed as

$$\omega\left(t_k\right)=v\left(t_k\right)\kappa\left(t_k\right). \tag{3}$$

iv) Turning angle, $\theta(\mathrm{t_k})$, given by

$$\left(t_k\right)=\tan^{-1}\left[\frac{\left(y_s\left(t_{k+1}\right)-y_s\left(t_k\right)\right)}{x_s\left(t_{k+1}\right)-x_s\left(t_k\right)}\right]. \tag{4}$$

v) Distance to track center, $r_C(t_k)$; provided that a geometrical center can be estimated for each track, $(x_c, y_c)$ (as the average coordinates in $x$ and $y$), then $r_C(t_k)$ is given by the distance of each track point $(x_s(\mathrm{t_k}), y_s(\mathrm{t_k}))$ from the corresponding $(x_c, y_c)$.

vi) Distance to cluster center, $d_C$; provided that a geometrical center of the entire cluster can be extracted $(\mathrm{X_c}, \mathrm{Y_c})$ as the average coordinates of all the track centers, then, $d_C$, is given by the average Euclidean distance between $(\mathrm{X_c}, \mathrm{Y_c})$ and all the $(x_c, y_c)$s.

vii) Diffusion coefficient, $D_2$,

$$D_2 = 4^{-1}e^{y_0}$$

with $y_0$ the y-axis intercept estimated from the Mean Square Displacement sequence [19].

viii) Directional persistence, $P$, given by the ratio of the distance between the starting and the ending point of the track ($L_0$ in panel (C) Fig. 1) and the actual length of the track, $L$.

Parameters i)-v) are time-varying descriptors. For this reason, we decided to extract aggregate statistical operators such as the average value, the standard deviation, the skewness, the kurtosis and the Shannon entropy for each parameter over the values collected along the track. In this way, not only the average value of the parameter, e.g., of the tangential speed magnitude, was computed but also high-order statistical moments. Such further descriptors accounted for the variation of the parameter along the track (standard deviation), the way this variation occurs (skewness and kurtosis), and the amount of information the distribution of values of the descriptor carries on (Shannon's Entropy). By merging the five statistical descriptors of parameters i)—v) (25 parameters) with constant parameters vi)-viii), a set of 28 descriptors was computed. Moreover, under the assumption of a consistent difference in short-term and long-term changes in track motility, as also demonstrated in [10], in presence of cell treatment, we derived two distinct sets of features: a first one computed over the entire observation time window (6 hours) and a second one computed on a limited time interval within the first 3 hours. Globally, we collected 56 different features of cell motility. This strategy was used in order to verify the presence of an eventual time-varying drug effect on the collective motion.

***3) Differential Analysis of Tracks Within Cluster:*** Cells do not react to drug in the same way, due to different factors, not least, forces of attraction and repulsion, adhesion, and biochemical signaling [10]. According to this, we observed that cells that lie in the central region of the cluster present a strongly confined movement thus reducing their ability to carry on the biological message of drug efficacy. In other words, whichever is their biological role, already to investigate, they do not adequately modify their migration behaviour in presence of increasing drug concentration. For this reason, we performed a distance-to-cluster-center dependent modeling of the phenomenon, by eliminating tracks whose center $(x_c, y_c)$ was too close to the cluster center $(\mathrm{X_c}, \mathrm{Y_c})$. By increasing this cut-off distance in a given range, we found improved results for a cut-off distance equal to about the average cancer cell diameter, as will be shown in Section IV in strong analogy to what found in the analysis carried on in [21].

***4) Classification Model: Track vs. Cluster:*** As shown in Fig. 1 and mentioned in Section II.B, after descriptors were collected, each track is classified using Linear Discriminant Analysis (LDA) model. LDA model assigns to each track one out of the three labels, 1 to 3, as shown in panel (D) in Fig. 1. However, under the assumption that a cluster of cells carries on a unified biological message related to drug efficacy, and

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TABLE I
SUMMARY OF THE AVAILABLE CASES. FROM LEFT TO RIGHT: EXPERIMENT IDENTIFICATION, DRUG CONCENTRATION ($\mu$M), NUMBER OF CLUSTERS, NUMBER OF TRACKS, ASSIGNED LABELS, AVERAGE CLUSTER AREA, AVERAGE NUMBER OF CELLS PER CLUSTER

| ID | Conc. [μM] | Clusters | Tracks | Label | Avg Area (μm²) | Avg no. Cells/clust |
|---|---|---|---|---|---|---|
| 1 | 0.5 | 6 | 192 | 2 | 1.61e+04 | 60 |
| 2 | 0.0 | 6 | 92 | 1 | 3.58e+03 | 11 |
| 3 | 0.0 | 5 | 145 | 1 | 7.92e+0.3 | 21 |
| 4 | 0.5 | 6 | 102 | 2 | 4.46e+03 | 13 |
| 5 | 5.0 | 5 | 177 | 2 | 7.68e+03 | 25 |
| 6 | 50.0 | 2 | 77 | 3 | 2.75e+03 | 9 |
| 7 | 5.0 | 5 | 160 | 2 | 1.13e+04 | 40 |
| 8 | 50.0 | 1 | 13 | 3 | 3.17e+03 | 7 |
| | | **36** | **958** | | | |

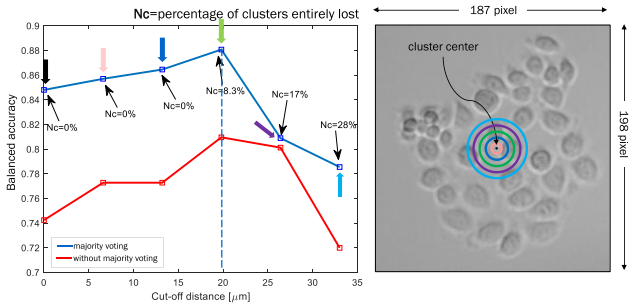



Fig. 2. (Left) Balanced accuracy of cluster classification vs. cut-off distance in the range [0–33]$\mu$m. The percentage of clusters ***Nc*** entirely eliminated from the analysis for increasing cut-off distance is also indicated; (right) correspondingly, colored circles at the cut-off distances are also plotted over a cluster of example. Blue and red curve refer to the application or not of the majority voting within the cluster.

according to the considerations reported in the Section I.B, we computed a second step in which the majority class assigned to the tracks in the same cluster was associated to the entire cluster (end part of panel (D) Fig. 1).

## III. EXPERIMENTAL SET-UP

The set of eight experiments listed in Table I has been considered in this work.

The settings correspond to four different experimental conditions in terms of drug concentration. From preliminary analysis, we discovered negligible differences in cell motility when concentrations 0.5 $\mu$M and 5.0 $\mu$M were administered. Consequently, we decided to finally assign the following labels: label $= 1$ to control experiments (untreated cells), label $= 2$ to cells treated with either 0.5 $\mu$M and 5.0 $\mu$M. Finally, a label $= 3$ is assigned to high concentrations (50 $\mu$M). A different number of tracks has been then processed, 237 for label $= 1$, 631 for label $= 2$, and 90 for label $= 3$. Simulations for classification model training were run using a leave-one-cluster-out cross-validation procedure. For ANOVA test, p-values less than 0.05 were considered significant. Bonferroni's correction will be applied to multi-comparison procedures.

## IV. RESULTS AND DISCUSSION

In order to demonstrate the effectiveness of the proposed platform, we apply the whole methodology to the 958 tracks reported in Table I. First, we verified the effect of the cut-off distance, that may lead to a diverse elimination of tracks close to cluster center, over classification accuracy (Section IV A), demonstrating the existence of an optimal cut-off distance for the analysis. Then, we evaluated accuracy in the three-class problem (labels 1–3) at the optimal cut-off distance of 30 px ($\sim$20 $\mu$m) (Section IV.B). Lastly, we also compared the distribution of some key descriptors and we performed ANOVA with Bonferroni's correction for multiple comparisons [30] in order to verify the statistical significance of the difference highlighted (Section IV.C).

### A. Effect of Cut-Off Distance Over Classification Accuracy

The cut-off distance determines the distance from the cluster center and define ***c*** cluster cores within which cell tracks are not considered for the analysis. Such a distance has to be chosen as a trade-off between number of tracks eliminated and performance of classification. The underlying assumption is that cells close to the center of a cluster move less than cells in the annular region of the cluster, and even less than those belonging to the rim region of the cluster [10], [21]. However, the latter cells may also exhibit misleading attitudes such as for example getting away along the radial direction, or simply belonging to portion of different clusters marginally visualized in the selected crop. For this reason, we implemented two automatic track selection criteria. Denoting as $(x_c, y_c)_i$ the track center of $i$-th cell and with $(X_c, Y_c)$ the cluster center to which it belongs, indicate with $r_i$ the Euclidean distance between $(x_c, y_c)_i$ and $(X_c, Y_c)$, and denote with $th_d$ the cut-off distance, we get the following condition: *eliminate tracks for which* $\boldsymbol{r}_i >$ quantile $(\boldsymbol{r}_i, 0.8)$ *(misleading tracks) or* $\boldsymbol{r}_i < \boldsymbol{th}_d$ *(tracks close to the center).* Simulations were performed by varying $\boldsymbol{th}_d$ in the range [0÷50]px (corresponding to [0÷33 $\mu$m]) with a step size equal to 10 px.

At each iteration, a leave-one-cluster-out procedure has been implemented. As the outcome of the cross-validation procedure, a label is assigned to all the tracks. By grouping the labels of the tracks of the same cluster in a whole via *cluster-oriented majority voting procedure*, each cluster is then labelled and balanced accuracy is computed. Clearly, the higher $\boldsymbol{th}_d$ is, the higher is the number of clusters entirely lost for the analysis (all the tracks within the small cluster are eliminated according to the condition defined above. Hence, Fig. 2 illustrates balanced accuracy achieved vs. the cut-off distance, by indicating also the percentage of clusters definitely lost in the analysis. The blue and the red curves indicate the accuracy trend with and without the application of the majority voting approach highlighting the superior performances of the majority strategy for all the considered cases.

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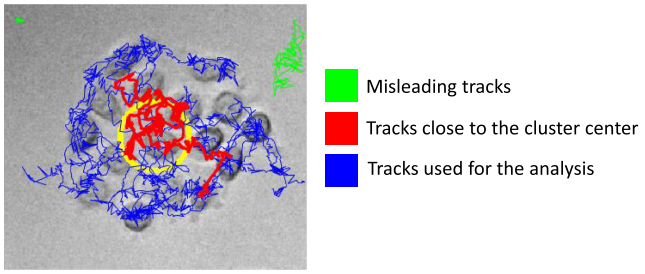



Fig. 3. Examples of track selection procedure. Green tracks represent misleading track far from the cluster center; red tracks have a cut-off distance smaller than the cut-off distance (yellow circle). Blue tracks are kept for further analysis.

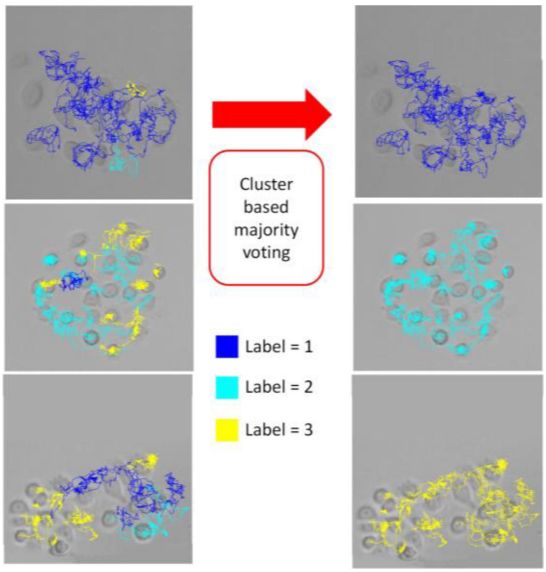



Fig. 4. Three examples of track classification (left column) vs. cluster classification (right column) using cluster-based majority voting procedure. Blue color indicates tracks assigned to label = 1, cyan tracks are the tracks assigned to label = 2, and yellow tracks are assigned to class labeled as 3.

Note that accuracy is clearly influenced by the selection of the cut-off distance especially for high $th_d$ values. Moreover, as expected the number of clusters lost, *Nc*, unacceptably increases thus leading to the optimal solution $th_d = 30$px ($\sim$20 $\mu$m) for both the classification strategies. Correspondingly, Fig. 3 illustrates a pictorial example of the tracks selection procedure for the majority voting with respect to the optimal cut-off distance. The yellow circle represents the selected cut-off distance. As expected tracks at the rim border of the cluster (green tracks) are kept off, the tracks whose centers fall within the cut-off distance (red tracks) are eliminated, while the remaining tracks (blue ones) are kept for the analysis.

### *B. Three-Class Cluster Classification Accuracy at Optimal Cut-Off Distance*

As it has been shown in Section IV.A, the optimal cut-off distance at which it is opportune to work is equal to 30 px ($\sim$20 $\mu$m), that corresponds to about the radius of cells involved (19 $\mu$m). In Fig. 4 we illustrate three examples of the final cluster labelling achieved. In the left column, the results provided by LDA model are shown whereas in the right column it is represented the result after majority voting.

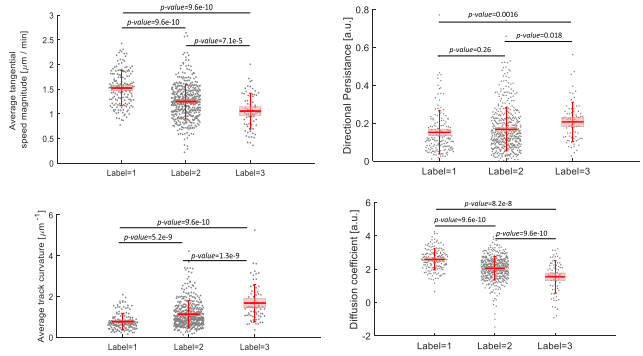



Fig. 5. Box-plots of the average tangential speed magnitude (upper-left), of the average track curvature (lower-left), of the directional persistence (upper-right), and of diffusion coefficient (lower-right) for tracks belonging to the three classes (Labels 1–3). The p-values of multi-comparison ANOVA test with Bonferroni's correction are also highlighted.

It can be noted that annular tracks mostly carry on the drug efficacy message and hence are fundamental for the final voting result. Blue tracks represent tracks assigned to class 1, cyan tracks to class 2, and yellow tracks to class 3. Accuracy values obtained in the experiment were equal to 83%, 81%, and 100% for clusters labelled as 1, 2, and 3 respectively. Remarkable is the fact that clusters targeted as 3 (high drug concentration) are completely recognized (accuracy = 100%). On the other hand, some clusters belonging to class labelled as 2 are assigned in turns to different classes (accuracy = 81%). On average, performance reaches an accuracy value of 88%.

### *C. Drug Concentration Induces Statistical Significance Difference in Kinematics*

Among the descriptors used in the presented multivariate analysis, some can be used to give an overall compact signature of the biological phenomenon under study. In particular, in Fig. 5, we represent the boxplots of the average tangential speed magnitude (upper-left), of the average track curvature (lower-right), of the directional persistence (upper-right), and of the diffusion coefficient (lower-right). For the sake of representation, only the interquartile range is shown (e.g., 25th–75th percentiles of the values are represented by the red shaded bar). The p-values of multi-comparison ANOVA test with Bonferroni's corrections are also indicated. As expected, speed magnitude decreases for higher drug concentration, thus indicating a primary drug effect of motion inhibition. On the other hand, the kind of the residual movements of the treated cells with respect to the untreated case present kinematics descriptors. As an example, the average curvature of the tracks increases for higher drug concentration meaning a more entropic, yet slower, motility (i.e., more confined tracks with the local smaller radius of curvature – high curvature values). At the same time, cells exposed to higher drug concentrations exhibit higher directional persistence and lower diffusion coefficient, confirming again a more confined motion.

Obviously, the fact that p-values exhibit mutually significant values, is only evidence of pairwise different feature distribution

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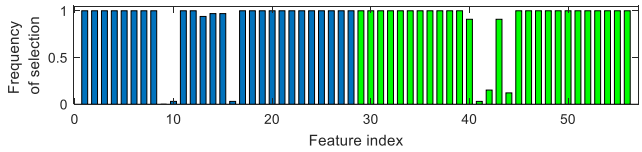



Fig. 6. Frequency of selection of the 56 features. Blue bars identify features collected over the first 3 hours while the green bars represent features collected over 6 hours.

but not directly a sign of class separability. Multivariate analysis is hence necessary to effectively assign samples to each class as previously shown. Another interesting aspect coming from the analysis of the features is highlighted observing the feature selection results performed in the leave-one-cluster-out cross-validation procedure. Fig. 6 illustrates the frequency of selection at each cluster-out by exploiting Fisher feature selection criterion [31]. Blue bars identify the 28 features collected over the first 3 hours while the green bars identify those collected over the 6 hours of experiments. The high selection rate of a large number of different features is a further confirmation of the need for multivariate analysis with respect to single feature decision-making strategy. Previous results demonstrate the existence of a deep complexity behind coordinated motility in terms of kind of motion, majority motility mode, and cell role within the same cluster. The multivariate analysis allows addressing this complexity by combining a set of kinematics descriptors able to represent the different facets of the phenomenon under investigation. Furthermore, the simultaneous selection of features collected during different temporal lags (3 and 6 hrs) provides a sketch of the presence of a time-varying drug effect on the collective motion. Future research work will address this aspect more in deep.

## V. Conclusion

Even though it is a key feature of cancer malignancy, cell movement has not been studied in deep as other phenomena, probably because of lack of proper experimental protocol. To better investigate this aspect, in this work we studied what happened to cell movement if we blocked cell replication. In particular, we investigated the effect of a chemotherapeutic agent (etoposide) causing a dose-dependent block of cell proliferation in the S phase of the cell cycle. Exploiting a synergic approach of cell tracking and multivariate data analysis we observed that, although the single cells maintain unique movement, they are also able to promote at the same time a majority behavior that is correlated with the different biological conditions. An extensive application of this methodology could give a different point of view of the cancer mechanisms.

## References

[1] E. Ben-Jacob and L. Herbert, "Self-engineering capabilities of bacteria," *J. Roy. Soc. Interface*, vol. 3, no. 6, pp. 197–214, 2006.
[2] T. J. Shaw and P. Martin, "Wound repair at a glance," *J. Cell Sci.*, vol. 122, no. 18, pp. 3209–3213, 2009.
[3] M. Poujade *et al.*, "Collective migration of an epithelial monolayer in response to a model wound," *Proc. Nat. Acad. Sci. USA*, vol. 104, no. 41, pp. 15988–15993, 2007.
[4] P. Friedl and D. Gilmour, "Collective cell migration in morphogenesis, regeneration and cancer," *Nature Rev. Mol. Cell Biol.*, vol. 10, no. 7, pp. 445–457, 2009.
[5] B. A. Camley *et al.*, "Emergent collective chemotaxis without single-cell gradient sensing," *Phys. Rev. Lett.*, vol. 116, no. 9, 2016, Art. no. 098101.
[6] A. J. Kabla, "Collective cell migration: Leadership, invasion and segregation," *J. Roy. Soc. Interface*, vol. 9, pp. 3268–3278, 2012.
[7] B. A. Camley and W.-J. Rappel, "Physical models of collective cell motility: From cell to tissue," *J. Phys. D, Appl. Phys.*, vol. 50, no. 11, 2017, Art. no. 113002.
[8] B. A. Camley *et al.*, "Polarity mechanisms such as contact inhibition of locomotion regulate persistent rotational motion of mammalian cells on micropatterns," *Proc. Nat. Acad. Sci. USA*, vol. 111, no. 41, pp. 14770–14775, 2014.
[9] F. Graner and J. A. Glazier, "Simulation of biological cell sorting using a two-dimensional extended Potts model," *Phys. Rev. Lett.*, vol. 69, no. 13, pp. 2013–2016, 1992.
[10] G. Malet-Engra *et al.*, "Collective cell motility promotes chemotactic prowess and resistance to chemorepulsion," *Current Biol.*, vol. 25, no. 2, pp. 242–250, 2015.
[11] S. Parlato *et al.*, "3D microfluidic model for evaluating immunotherapy efficacy by tracking dendritic cell behaviour toward tumor cells," *Sci. Rep.*, vol. 7, 2017, Art. no. 1093.
[12] E. Biselli *et al.*, "Organs on chip approach: A tool to evaluate cancer-immune cells interactions," *Sci. Rep.*, vol. 7, no. 1, 2017, Art. no. 12737.
[13] J.-Y. Tinevez *et al.*, "TrackMate: An open and extensible platform for single-particle tracking," *Methods*, vol. 115, pp. 80–90, 2017.
[14] E. Meijering *et al.*, "Methods for cell and particle tracking," in *Methods in Enzymology*, vol. 504. New York, NY, USA: Academic, 2012, pp. 183–200.
[15] N. Chenouard *et al.*, "Objective comparison of particle tracking methods," *Nature Methods*, vol. 11, no. 3, pp. 281–289, 2014.
[16] R. Gorelik and A. Gautreau, "Quantitative and unbiased analysis of directional persistence in cell migration," *Nature Protocols*, vol. 9, no. 8, pp. 1931–1943, 2014.
[17] J. Huth *et al.*, "Significantly improved precision of cell migration analysis in time-lapse video microscopy through use of a fully automated tracking system," *BMC Cell Biol.*, vol. 11, no. 1, pp. 1–12, 2010.
[18] D. A. Chapnick *et al.*, "The development of a novel high throughput computational tool for studying individual and collective cellular migration," *PloS One*, vol. 8, no. 12, 2013, Art. no. e82444.
[19] I. F. Sbalzarini and P. Koumoutsakos, "Feature point tracking and trajectory analysis for video imaging in cell biology," *J. Struct. Biol.*, vol. 151, no. 2, pp. 182–195, 2005.
[20] M. Deforet *et al.*, "Automated velocity mapping of migrating cell populations (AVeMap)," *Nature Methods*, vol. 9, no. 11, pp. 1081–1083, 2012.
[21] F. J. Segerer *et al.*, "Emergence and persistence of collective cell migration on small circular micropatterns," *Phys. Rev. Lett.*, vol. 114, no. 22, 2015, Art. no. 228102.
[22] W. Geng *et al.*, "Automatic tracking, feature extraction and classification of C. elegans phenotypes," *IEEE Trans. Biomed. Eng.*, vol. 51, no. 10, pp. 1811–1820, Oct. 2004.
[23] J. Xie *et al.*, "Automatic tracking of Escherichia coli in phase-contrast microscopy video," *IEEE Trans. Biomed. Eng.*, vol. 56, no. 2, pp. 390–399, Feb. 2009.
[24] M. Dewan *et al.*, "Tracking biological cells in time-lapse microscopy: An adaptive technique combining motion and topological features," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 6, pp. 1637–1647, Jun. 2011.
[25] K. G. Murthy, *Network Programming*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1992.
[26] J. Munkres, "Algorithms for the assignment and transportation problems," *J. Soc. Ind. Appl. Math.*, vol. 5, no. 1, pp. 32–38, 1957.
[27] X. Chen *et al.*, "Automated segmentation, classification, and tracking of cancer cell nuclei in time-lapse microscopy," *IEEE Trans. Biomed. Eng.*, vol. 53, no. 4, pp. 762–766, Apr. 2006.
[28] E. R. Davies, *Machine Vision: Theory, Algorithms, Practicalities*. New York, NY, USA: Elsevier, 2004.
[29] H. K. Yuen *et al.*, "Comparative study of Hough transform methods for circle finding," *Image Vis. Comput.*, vol. 8, no. 1, pp. 71–77, 1990.
[30] R. J. Cabin and R. J. Mitchell, "To Bonferroni or not to Bonferroni: When and how are the questions," *Bull. Ecol. Soc. Amer.*, vol. 81, no. 3, pp. 246–248, 2000.
[31] G. Magna *et al.*, "Unsupervised on-line selection of training features for a robust classification with drifting and faulty gas sensors," *Sensors Actuators B, Chem.*, vol. 258, pp. 1242–1251, 2018.

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